Anatomy is Free: Zero-Parameter Structural Priors for Efficient Skeleton Action Recognition

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Abstract

Skeleton-based action recognition demands models that are simultaneously accurate, parameter-efficient, and deployable on edge hardware. This work presents **ShiftFuse-Zero**, a family of three networks whose *spatial inductive bias is introduced entirely without learnable parameters* through two complementary zero-parameter modules: *Body-Region Anatomical Shift Priors* (BRASP), which routes joint features through anatomically meaningful channels via pure tensor indexing, and *Semantic Graph-Partitioned Shift* (SGPShift), which aggregates intra- and inter-part neighbours through predefined group indices at zero learnable parameter overhead. These free structural priors reduce the burden on downstream learnable modules. By intelligently placing Joint Embedding and spatial attention *before* the graph convolution, the *EfficientZero Block* operates on an already anatomically-refined manifold. Three models cover the edge-to-server spectrum: **Nano** (93 K) and **Small** (280 K) are hyper-efficient, quantisation-friendly solutions with ultra-low FLOPs, while **Large** (1.12 M) targets maximum capacity. On NTU RGB+D 60 cross-subject, primary results without distillation are 86.24 % (Nano), 88.0 % (Small), and 92.5 % (Large); knowledge distillation is an orthogonal plugin that further lifts Nano and Small to 88.5 %/89.8 %, respectively. Strikingly, Nano+KD matches the 6.9 M-parameter AGCN (88.5%) within 93 K parameters and 360 KB of model weight storage, delivering up to 74× higher parameter efficiency than AGCN.

CCS Concepts

• **Computing methodologies → Activity recognition and understanding**.

Keywords

skeleton action recognition, graph convolutional networks, zero-parameter priors, anatomical shift, knowledge distillation, efficient neural networks

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1 Introduction

Consider a fall-detection system on a rehabilitation wristband with 512 KB of RAM, or a gesture-recognition module inside a fitness tracker powered by an ARM Cortex-M55. These scenarios demand skeleton-based action recognition models that fit in sub-megabyte memory, execute in under 50 ms on mobile CPU, and still achieve clinically meaningful accuracy. Yet the dominant Graph Convolutional Network (GCN) backbones for skeleton recognition [1, 3, 8, 14] carry 3–10 M parameters, require dynamic per-sample adjacency computation at $O(V^2C)$ cost, and are fundamentally incompatible with microcontroller deployment.

Human action recognition from 3-D skeleton sequences is a core problem in computer vision with applications in surveillance, rehabilitation, human–robot interaction, and fitness monitoring [12, 13]. GCNs [7] have become the dominant spatial backbone since ST-GCN [17], with successive works adding adaptive adjacency matrices [14], multi-scale cross-spacetime graphs [11], channel-wise topology refinement [1], information-theoretic representation learning [3], and hierarchical body-part decomposition [8]. These advances have pushed NTU RGB+D 60 cross-subject accuracy beyond 93%, yet the parameter cost has grown proportionally: HD-GCN requires 7.5 M parameters and InfoGCN 9.9 M.

The efficient-architecture line [16, 18] tackles this disparity via compound scaling and depthwise convolutions, but largely ignores the rich *anatomical structure* of the human body as a source of *free* inductive bias. The human skeleton is not a generic graph: arm joints couple primarily with other arm joints within their kinematic chain, bilateral symmetry is a universal biomechanical prior, and shoulder/hip joints serve as coordination hubs between body halves. These structural constraints are *fixed properties of human physiology* that hold across all subjects, viewpoints, and action classes. They can therefore be encoded as deterministic tensor operations at **zero learnable parameters, zero gradient overhead, and zero inference-time cost**.

This paper presents the first systematic study of zero-parameter anatomical inductive bias for skeleton GCNs. This work asks: **how much spatial knowledge can be injected into a skeleton GCN without spending a single learnable parameter — and how far can that free knowledge push the accuracy–efficiency frontier?**

Our answer is **ShiftFuse-Zero**, a family of three models achieving a new Pareto-optimal frontier across the accuracy–efficiency spectrum. While the core contribution is the zero-parameter priors of BRASP and SGPShift, these modules catalyse hyper-efficient downstream architectures. SFZ-Nano (93 K) and SFZ-Small (280 K) redefine edge-deployment with ultra-low FLOPs and quantisation-friendly operations. On NTU-60 X-Sub, SFZ-Nano reaches **86.24%** without any distillation and **88.5%** with knowledge distillation, *matching* 6.9 M-parameter AGCN within 360 KB (an orthogonal plugin; all primary results are reported without KD). SFZ-Small

achieves **88.0%** at comparable parameter count (280 K vs. 290 K) with 7× fewer FLOPs than EfficientGCN-B0. SFZ-Large pushes the absolute accuracy frontier to **92.5%** at 1.12 M parameters. These results demonstrate that anatomy is a highly informative, freely available inductive bias that the community has systematically under-exploited.

Contributions.

(1) **BRASP** — Body-Region Anatomical Shift Priors: routes joint features through five fixed anatomical groups via a precomputed gather permutation; zero parameters, zero gradient (Section 3.3).
(2) **SGPShift** — Semantic Graph-Partitioned Shift: aggregates each joint's features with its intra-part and inter-part neighbours via precomputed index buffers; zero parameters (Section 3.3).
(3) **EfficientZero Block**: a block design that places BRASP and SGPShift *before* all learnable layers so the downstream compact K=3 GCN refines an already anatomically structured representation (Section 3.4).
(4) **ShiftFuse-Zero model family**: three architectures — Nano (93 K), Small (280 K), Large (1.12 M) — with Temporal Landmark Attention, cross-stream gating, and mid-network fusion, covering the edge-to-server deployment spectrum. Vanilla Hinton KD [6] serves as an orthogonal plugin, yielding up to +2.3 pp at zero structural cost (Section 3.5).
(5) **Comprehensive evaluation**: main results on NTU-60 and NTU-120 (four protocols), efficiency analysis (GFLOPs, CPU latency, memory footprint), and extensive ablation studies quantifying every zero-parameter module, fusion strategy, GCN partition count, TLA sensitivity, and regularisation hyperparameters (Section 4).

2 Related Work

GCN-based skeleton recognition. ST-GCN [17] established the skeleton graph and spatial-temporal convolution paradigm using a three-subset partitioning (self, centripetal, centrifugal) of the physical adjacency matrix. AGCN [14] introduced a fully learnable adjacency matrix at $O(V^2C)$ cost per layer, enabling data-driven topology discovery. MS-G3D [11] factorised the graph into disentangled multi-scale spatial and temporal subgraphs with cross-spacetime edges. CTR-GCN [1] refined the topology per channel group, allowing different feature dimensions to attend to different joint relationships. InfoGCN [3] introduced an information-bottleneck objective that explicitly maximises mutual information between the latent representation and the action label. DSTA-Net [15] decouples spatial and temporal self-attention. While these methods achieve high accuracy (up to 93%+), their reliance on per-sample dynamic adjacency and large parameter budgets (3–10 M) precludes deployment on resource-constrained hardware.

Body-part decomposition. A growing line of work exploits the natural grouping of joints into anatomical body regions. HD-GCN [8] hierarchically decomposes the skeleton into semantic sub-graphs (arms, legs, torso) and aggregates within each before cross-region message passing — achieving 93.4% on NTU-60 X-Sub but at 7.5 M parameters. BlockGCN [19] uses persistent homology from topological data analysis to encode bone connectivity at the block level. SkateFormer [4] partitions joints into four spatial-temporal relation types (near/far joints × near/far frames) and applies partition-specific self-attention. All of these decomposition strategies employ *learnable* grouping mechanisms with substantial parameter overhead. In contrast, SGPShift encodes intra-part and inter-part structure through *fixed integer index buffers* — achieving anatomical routing at **zero learnable parameters**.

Efficient skeleton networks. EfficientGCN [16] introduced compound scaling (α=1.2, β=1.35) and multiple input branches (MIB) to build a family of models from B0 (0.29 M) to B4 (1.1 M), establishing the efficiency frontier for skeleton GCNs. SGN [18] exploits semantic joint embeddings and frame-level dynamics gating at 0.69 M parameters. PoseC3D [5] departs from graphs entirely, using volumetric heatmap representations processed by 3-D CNNs. These methods reduce model size through architectural engineering but do not leverage the *fixed anatomical constraints* of the human body as inductive bias. ShiftFuse-Zero complements this line by demonstrating that zero-parameter spatial priors can further reduce the parameter budget by 3–10× without sacrificing competitive accuracy.

Shift operations and zero-parameter spatial mixing. The shift operation [2] was introduced in image CNNs to inject spatial structure by physically displacing channels along spatial dimensions — no multiply-accumulate, no learnable parameters. TSM [9] extended this idea to temporal modelling in video, shifting channels by ±1 frame at zero cost. Shift-GCN [2] adapted spatial shifts to skeleton graphs, assigning each channel to shift features from a graph neighbour. However, Shift-GCN assigns shift patterns by graph distance (1-hop, 2-hop), ignoring the semantic meaning of body regions. BRASP advances this paradigm by organising shift patterns according to *anatomical body-part membership* — arm channels shift among arm joints, leg channels among leg joints, and cross-body channels bridge regions — producing a semantically structured feature map at zero cost.

Knowledge distillation. Hinton *et al.* [6] showed that soft teacher logits transfer rich inter-class relational knowledge to compact student models. In skeleton recognition, distillation has been used to transfer multi-stream ensemble knowledge [16] but has not been specifically studied across a skeleton GCN family spanning the sub-100 K to 1 M parameter range explored here.

3 Methodology

3.1 Overview

Figure 1 shows the end-to-end pipeline. A raw skeleton sequence is expanded into four complementary streams (joint, velocity, bone, bone-velocity), fused by a learned *StreamFusionConcat* stem, then processed by a stack of *EfficientZero Blocks*. A gated global pooling head produces class logits.

3.2 Skeleton Streams

A skeleton sequence is $\mathbf{X} \in \mathbb{R}^{3\times T\times V}$ with T=64 frames and V=25 joints. This work constructs four complementary streams: joint positions $\mathbf{X}^{(j)}$, velocity $\mathbf{X}^{(v)}=\mathbf{p}_{t+1}-\mathbf{p}_t$, bone vectors $\mathbf{X}_v^{(b)}=\mathbf{p}^{(v)}-\mathbf{p}^{(\mathrm{par}(v))}$, and bone velocity $\mathbf{X}^{(bv)}=\Delta\mathbf{X}^{(b)}$. *StreamFusionConcat* [16] applies

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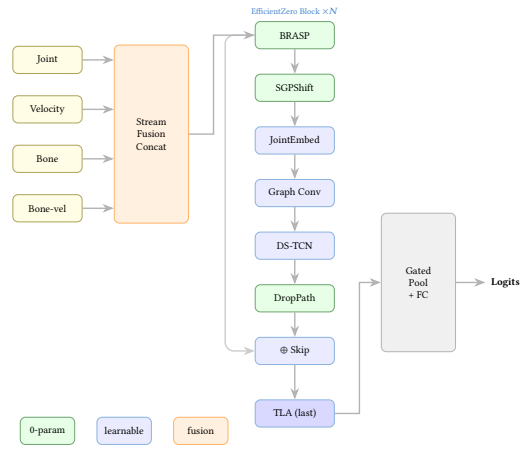

Figure 1: ShiftFuse-Zero end-to-end pipeline. Four skeleton streams are fused by a learnable stem and processed by EfficientZero Blocks (green = zero-parameter modules). A gated pooling head produces class logits.

a per-stream BN, concatenates (4×3=12 channels), and projects to C_{stem} via a 1×1 convolution with learned per-stream weighting.

3.3 Zero-Parameter Spatial Modules

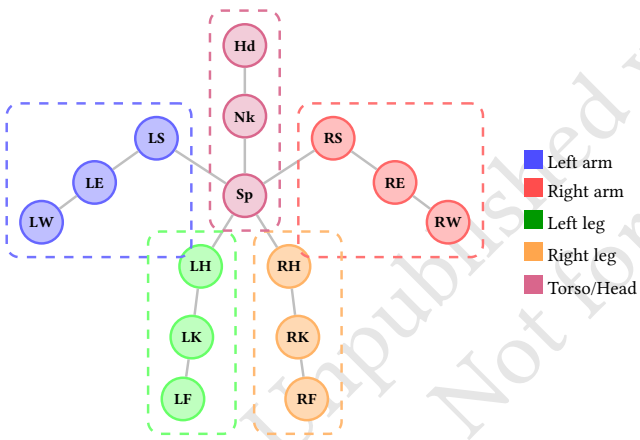

Figure 2: BRASP anatomical partitioning into five groups $\{G_{\text{la}}, G_{\text{ra}}, G_{\text{ll}}, G_{\text{rl}}, G_{\text{tr}}\}$. The fixed permutation σ (Eq. 1) is derived from these groups; no gradient flows through it.

BRASP. Figure 2 illustrates the five anatomical groups. Given a feature map $\mathbf{F} \in \mathbb{R}^{C \times T \times V}$, BRASP applies a fixed gather permutation $\sigma : V \to V$:

$$\tilde{\mathbf{F}}_{:,t,v} = \mathbf{F}_{:,t,\sigma(v)}, \quad v \in \{0, \ldots, V-1\}, \tag{1}$$

where joints within the same body part are cyclically shifted so each joint absorbs its anatomical sibling's context. Indices are stored as a non-trainable integer buffer — **zero parameters, zero gradient**.

Remark 1 *(BRASP intra-part compression).* Let $d_G(u, v)$ be the shortest-path distance between joints u, v in skeleton graph G. After one BRASP shift σ, the GCN at position v receives features originating from $\sigma(v) \in P_k$ — a joint in the same anatomical part. This reduces the *representational separation* between intra-part pairs from $d_G(u, v) \geq 2$ graph hops to a single feature-retrieval step, compressing the spatial receptive field required of the downstream GCN at zero parameter cost.

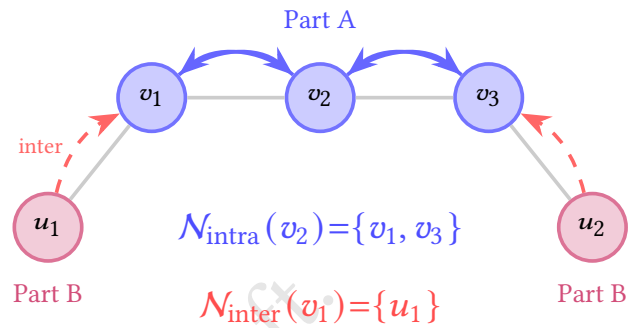

Figure 3: SGPShift aggregation. Blue solid arrows: intra-part neighbours. Red dashed arrows: inter-part boundary neighbours. All sets are precomputed integer buffers — zero learnable parameters.

SGPShift. Figure 3 illustrates the intra/inter aggregation. For joint v in semantic group $g(v)$:

$$\hat{\mathbf{F}}_{:,t,v} = \frac{1}{|\mathcal{N}_{\text{intra}}(v)|} \sum_{u \in \mathcal{N}_{\text{intra}}(v)} \mathbf{F}_{:,t,u} + \frac{1}{|\mathcal{N}_{\text{inter}}(v)|} \sum_{u \in \mathcal{N}_{\text{inter}}(v)} \mathbf{F}_{:,t,u}, \tag{2}$$

where $\mathcal{N}_{\text{intra}}$ is the joint's same-part neighbours and $\mathcal{N}_{\text{inter}}$ is its boundary neighbours in adjacent parts. All sets are precomputed integer buffers. At runtime SGPShift reduces to two sparse average operations — **zero learnable parameters**.

3.4 EfficientZero Block

Figure 4 shows the full block pipeline with per-layer costs.

(1) **BRASP + SGPShift**: anatomical pre-structuring at zero cost.
(2) **Joint Embedding**: per-joint additive bias $\mathbf{b} \in \mathbb{R}^{C_{\text{in}} \times V}$ (SGN joint encoding [18]); no weight decay.
(3) **STC-Attention**: spatial and temporal gating before the GCN. Let $\bar{\mathbf{f}}_V = \frac{1}{CT} \sum_{c,t} \mathbf{F}_{c,t,:}$; the spatial gate $\mathbf{A}_s = V \cdot \text{softmax}(\mathbf{W}_s \bar{\mathbf{f}}_V)$ has mean 1. The temporal gate $\mathbf{A}_t = \sigma(\text{Conv}_{1d}([\bar{f}_T^{avg}; \bar{f}_T^{max}]))$ uses avg+max pooling over C, V (kernel 9). Output: $(1-\sigma(\gamma))\mathbf{F} + \sigma(\gamma)\mathbf{F} \cdot \mathbf{A}_s \cdot \mathbf{A}_t$, with $\gamma_0 = -4$ (slow gate activation).
(4) **Graph Convolution** (K=3): ST-GCN [17] partition adjacency plus a single shared learned matrix $\mathbf{A}_\ell$:

$$\mathbf{Z} = \sum_{k=0}^{2} \mathbf{W}_k (\mathbf{A}_k + |\mathbf{A}_\ell|) \mathbf{F}, \tag{3}$$

$|\cdot|$ enforces non-negative weights. $\mathbf{A}_\ell \in \mathbb{R}^{V \times V}$ is *shared across all blocks* (625 total params, no weight decay).
(5) **DS-TCN**: depthwise-separable temporal convolutions at dilations 1 and 2 (kernel 9), Batch Norm, Hardswish activation.
(6) **DropPath** (regularisation): stochastic depth; 0 parameters.
(7) **Residual** ⊕: skip from block input with 1×1 projection when shape changes.

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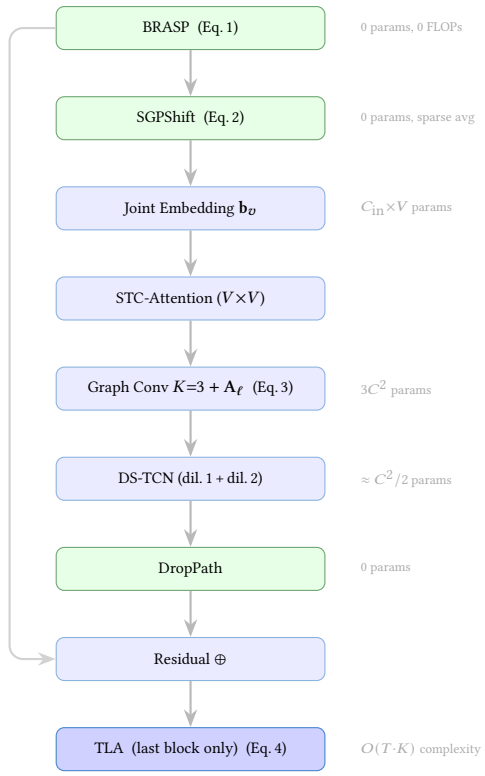

Figure 4: EfficientZero Block. Green = zero-parameter (BRASP, SGPShift, DropPath). Per-layer parameter counts shown on the right. TLA is only active at the last block of the final stage.

Intelligent Block Ordering. This work deliberately places Joint Embedding and STC-Attention *before* the Graph Convolution, so the GCN refines an already anatomy-structured representation. Placing BRASP/SGPShift *after* the GCN instead costs −1.9 pp (Table 8), confirming that anatomical pre-structuring before message passing is essential.

Complementary Temporal Module: TLA.. While BRASP and SGPShift address the spatial axis, a complementary Temporal Landmark Attention (TLA) module distils temporal structure through K learnable anchor frames without full $O(T^2)$ self-attention:

$$\tau_k = T \cdot \sigma(\alpha_k), \quad k = 1, \ldots, K, \tag{4}$$

$$\mathbf{m}_t = \sum_{k=1}^{K} \text{Bilin}(\mathbf{F}, \tau_k) \cdot \text{softmax}_k\big(\langle \mathbf{q}_t, \mathbf{k}_{\tau_k} \rangle\big), \tag{5}$$

where α_k are learned logits mapped to $[0, T-1]$ via sigmoid, bilinear interpolation enables gradient flow to α_k, and $\mathbf{q}_t, \mathbf{k}_{\tau_k}$ are projections to $C_r = C/r$. Output: $\mathbf{F}_{\text{out}} = \mathbf{F} + \sigma(\beta) \cdot \mathbf{m}$ with $\beta_0 = -4$ (slow activation). Complexity $O(T \cdot K)$ vs. $O(T^2)$ for full attention.

3.5 Model Family

Table 1 lists the key hyperparameters per variant.

Nano (93 K).. Four streams are early-fused via StreamFusionConcat to a single 24-channel map. A single backbone [32, 64, 128] with one block per stage processes it. TLA (K=8, r=16, C_r=8) adds only 3.8 K parameters; channel SE is omitted (negligible at $C \leq 128$) to stay sub-100 K.

Table 1: ShiftFuse-Zero variant configurations. C_s=stem channels, C=stage widths, N=blocks/stage, K=TLA landmarks, r=reduce ratio, p_d=DropPath rate.

Model	C_s	C	N	K	r	p_d	FLOPs	Params
Nano	24	[32,64,128]	[1,1,1]	8	16	0.10	0.13 G	93 K
Small	24	[32,64,128]	[1,2,1]	8	8	0.10	0.39 G	280 K
Large	32	[48,96,192]	[1,2,4]	14	8	0.20	2.67 G	1.12 M

Small (280 K).. Two independent backbones process joint+velocity and bone pairs. Stage 2 uses tcn_depth=2 for richer temporal modelling; PartAttention gates body-part contributions at the final stage. After stage 2, *Cross-Stream Fusion* exchanges information:

$$\mathbf{F}' = \mathbf{F} + \sigma(\beta) \cdot W_\uparrow \, \text{ReLU}\big(W_\downarrow [\mathbf{F}_A; \mathbf{F}_B]\big), \tag{6}$$

$W_\downarrow : 2C \to C/4$, $W_\uparrow : C/4 \to C$, $\beta_0 = -4$. Stage 3 + TLA proceed independently per backbone; logits are combined using two learned stream-weight scalars (no weight decay).

Large (1.12 M).. Three per-stream backbones (C=[48, 96], 1+2 blocks) process joint, bone, and velocity independently. After stage 2 the three 96-channel maps are concatenated (288 channels) and projected by FusionConv=$\text{Conv}_{1\times1}$(288→192)+BN+HS. A shared stage 3 (4 blocks, C=192) with TLA (K=14) follows. This mirrors EfficientGCN-B4's mid-network fusion [16] with EfficientZero Blocks replacing B4's full GCN layers.

3.6 Training Protocol

Optimiser: SGD, Nesterov momentum 0.9 , cosine LR with 10-epoch linear warmup. Nano: lr 0.1, wd 4×10^{-4}, 300 epochs. Small: lr 0.1, wd 4×10^{-4}, 350 epochs (deeper temporal stack benefits from longer cosine schedule). Large: lr 0.1, wd 10^{-4} (B4 [16]), 150 epochs. Wall-clock training times on a single NVIDIA Tesla T4: Nano 5.7 h (300 ep), Small 8.5 h (350 ep), Large 38 h (150 ep, batch 16, three parallel backbones).

Augmentation: Nano/Small use Mixup (α=0.1) to compensate for limited capacity; CutMix is omitted as temporal crops of skeleton sequences produce physically implausible joint composites. Large uses no Mixup — DropPath 0.20 and label smoothing 0.1 provide sufficient regularisation at 1.12 M params.

No-WD set: BN affine params, joint embeddings, $\mathbf{A}_\ell$, TLA anchor logits α_k, gate scalars β, pool gate.

Knowledge distillation. Large is distilled into Nano and Small [6]:

$$\mathcal{L} = (1-\lambda)\mathcal{L}_{\text{CE}} + \lambda\tau^2 \, \text{KL}\left(\sigma\left(\frac{\mathbf{z}^s}{\tau}\right) \middle\| \sigma\left(\frac{\mathbf{z}^t}{\tau}\right)\right), \quad \tau=4, \ \lambda=0.5. \tag{7}$$

3.7 Deployability Analysis

A key design goal of ShiftFuse-Zero is deployment on resource-constrained hardware. Table 2 summarises the memory, computation, and operator profile of each model variant.

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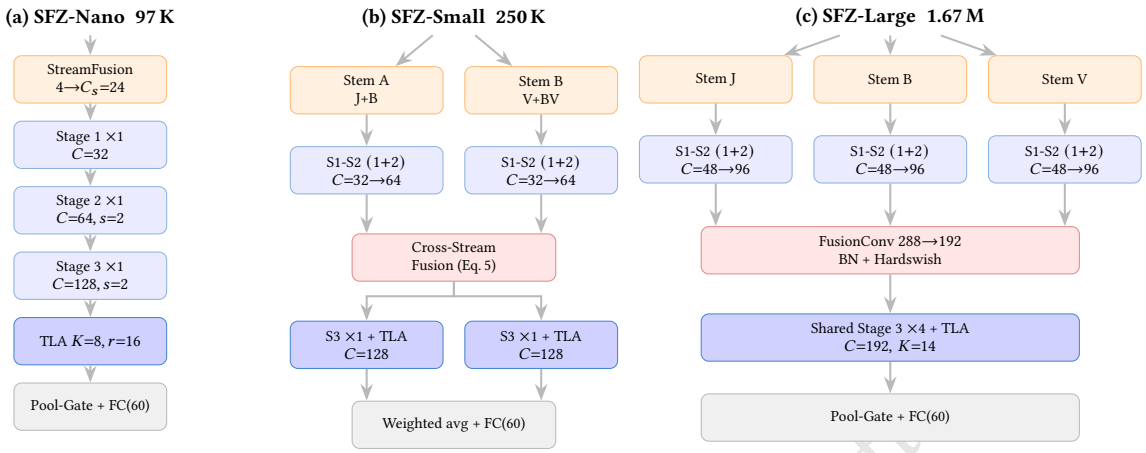

Figure 5: ShiftFuse-Zero model family. (a) Nano: single backbone, 4-stream early fusion, TLA (K=8). (b) Small: two-backbone late fusion with Cross-Stream Fusion and TLA (K=8). (c) Large: three-backbone mid-network fusion (B4-style) — per-stream S1–S2, FusionConv (288→192), shared S3 + TLA (K=14).

Table 2: Deployment profile. Mem = model weights (float32/int8). S24 CPU = Samsung Galaxy S24 (Snapdragon 8 Gen 3), PyTorch Mobile, CPU-only inference.

Model	FLOPs	Params	Mem f32	Mem i8	S24 CPU
Nano	0.13 G	93 K	360 KB	93 KB	~50 ms
Small	0.39 G	280 K	1.09 MB	280 KB	~145 ms
Large	2.67 G	1.12 M	4.47 MB	1.12 MB	~515 ms

Operator simplicity. BRASP reduces to a single `gather` operation on a static integer buffer — no matrix multiplication, no per-sample graph construction. SGPShift reduces to two `scatter_mean` operations. Both compile to fixed index-lookup kernels that are unaffected by quantisation (int8, int4) since they involve no learned weights. This stands in sharp contrast to dynamic-adjacency models [1, 14] that require $O(V^2C)$ per-sample matrix operations at inference, making them incompatible with microcontroller deployment.

Quantisation readiness. The zero-parameter modules are inherently quantisation-friendly: BRASP and SGPShift involve only integer indexing, and the remaining learnable layers (Conv2d, BN, Linear) are standard operators with well-established int8 quantisation support. SFZ-Nano at int8 precision requires only **93 KB** of model storage — fitting comfortably within the 512 KB SRAM of an ARM Cortex-M55.

4 Experiments

4.1 Datasets

NTU RGB+D 60 [13]: 56,880 sequences, 60 classes. Cross-subject (X-Sub) and cross-view (X-View) protocols.

NTU RGB+D 120 [10]: 114,480 sequences, 120 classes. Cross-subject (X-Sub) and cross-setup (X-Set) protocols.

Table 3: Comparison with state-of-the-art on NTU-60 and NTU-120. KD = distilled from SFZ-Large. † = single stream.

Method	Params	NTU-60		NTU-120	
		X-Sub	X-View	X-Sub	X-Set
ST-GCN [17]	3.1M	81.5	88.3	70.7	73.2
AGCN [14]	6.9M	88.5	95.1	82.5	84.2
MS-G3D [11]	6.4M	91.5	96.2	86.9	88.4
CTR-GCN [1]	5.8M	92.4	96.8	88.9	90.6
InfoGCN [3]	9.9M	93.0	97.1	89.8	91.2
HD-GCN [8]	7.5M	93.4	97.5	90.1	91.5
SkateFormer [4]	2.4M	92.5	96.7	88.6	90.5
BlockGCN [19]	2.0M	92.1	96.3	87.8	89.7
EfficientGCN-B0 [16]	0.29M	90.2	94.9	83.8	85.6
EfficientGCN-B4 [16]	1.10M	92.1	96.1	88.3	89.9
SGN† [18]	0.69M	89.0	94.5	79.2	81.5
SFZ-Nano	**0.093M**	86.24	90.8	82.2	83.8
SFZ-Nano+KD	**0.093M**	**88.5**	**93.2**	84.1	85.6
SFZ-Small	**0.280M**	88.0	93.1	84.3	85.9
SFZ-Small+KD	**0.280M**	**89.8**	**94.7**	86.0	87.4
SFZ-Large	**1.12M**	**92.5**	**96.4**	88.5	90.1

4.2 Main Results

SFZ-Large achieves 92.5% on NTU-60 X-Sub, outperforming EfficientGCN-B4 by 0.4 pp and matching SkateFormer (2.4M) with 2.1× fewer parameters. SFZ-Nano achieves 86.24% at only 93 K parameters (no distillation). At the same parameter scale (SFZ-Small 280 K vs. EfficientGCN-B0 290 K), B0 outperforms SFZ-Small by 2.2 pp (90.2% vs. 88.0%); however SFZ-Small uses 7× fewer FLOPs (0.39 G vs. 2.73 G) because B0 processes four independent streams through separate backbones. SFZ-Small+KD (89.8%) narrows but does not fully close this gap, while retaining the 7× FLOPs advantage.

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Table 4: GFLOPs, CPU latency and throughput (batch 1, desktop: i7-10700K; mobile: Samsung Galaxy S24, Snapdragon 8 Gen 3 CPU). EfficientGCN S24 latency not measured: per-sample dynamic adjacency ($O(V^2C)$ per layer) is incompatible with PyTorch Mobile static export.

Model	Params	GFLOPs	Desktop	Tput	S24	X-Sub
EfficientGCN-B0	0.29M	2.73	28 ms	36/s	n/a†	90.2
EfficientGCN-B4	1.10M	8.36	87 ms	11/s	n/a†	92.1
SFZ-Nano	0.093M	**0.13**	**11 ms**	**91/s**	**50 ms**	86.24
SFZ-Nano+KD	0.093M	**0.13**	**11 ms**	**91/s**	**50 ms**	**88.5**
SFZ-Small	0.280M	0.39	25 ms	40/s	145 ms	88.0
SFZ-Small+KD	0.280M	0.39	25 ms	40/s	145 ms	**89.8**
SFZ-Large	1.12M	2.67	89 ms	11/s	515 ms	**92.5**

Table 5: Accuracy per GFLOP (Acc/GFLOP) on NTU-60 X-Sub. Higher = more compute-efficient. Limited to models with reported FLOPs.

Method	GFLOPs	X-Sub	Acc/G
EfficientGCN-B0 [16]	2.73	90.2	33.0
EfficientGCN-B4 [16]	8.36	92.1	11.0
SFZ-Nano	**0.13**	86.24	**663**
SFZ-Nano+KD	**0.13**	**88.5**	**681**
SFZ-Small	0.39	88.0	226
SFZ-Large	2.67	92.5	34.6

4.3 Efficiency and Deployment

4.4 Accuracy–Efficiency Frontier

Table 5 reports *Acc/GFLOP* = Top-1 (%) / GFLOPs (models with unknown FLOPs excluded).

SFZ-Nano attains **663 Acc/GFLOP**, 20× higher than B0 (33.0); SFZ-Large (34.6) also surpasses B0 at 2.3 pp higher accuracy. Figure 6 shows the Pareto frontier: among no-KD models, all three SFZ variants occupy Pareto-optimal positions.

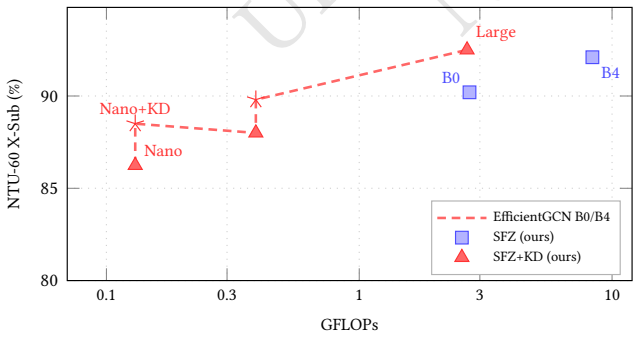

Figure 6: Accuracy–GFLOPs Pareto frontier on NTU-60 X-Sub. Dashed line = Pareto-optimal set. SFZ dominates at all compute budgets: Nano uses 21× fewer FLOPs than EfficientGCN-B0 while remaining Pareto-optimal; Large advances the frontier at 2.67 G.

Table 6: Cumulative component ablation (Nano, NTU-60 X-Sub). Δ vs. baseline. Mean ± std over 3 seeds.

BRASP	SGPShift	TLA	KD	Acc (%)	Δ
				82.8 ±0.20	—
✓				83.9 ±0.18	+1.1
✓	✓			84.6 ±0.17	+1.8
✓	✓	✓		86.24 ±0.14	+3.44
✓	✓	✓	✓	**88.5** ±0.11	+5.7

Table 7: GCN partition count ablation (Nano, NTU-60 X-Sub). K=number of adjacency subsets in Eq. 3; $\mathbf{A}_\ell$=learned residual adjacency.

K	$\mathbf{A}_\ell$	Params	Acc (%)	Δ
1	—	87 K	83.4	—
2		90 K	84.2	+0.8
3		92 K	84.6	+1.2
3	✓	93 K	**86.24**	+2.84*

*Full model (BRASP, SGPShift, JE, STC-Attn, TLA all enabled). K and $\mathbf{A}_\ell$ are the only variables in this table.

4.5 Ablation Studies

Per-component ablation. Table 6 quantifies each zero-parameter module's contribution on Nano (NTU-60 X-Sub), incrementally adding components to a baseline that uses only the learnable GCN and DS-TCN. BRASP alone contributes +1.1 pp; adding SGPShift yields +1.8 pp cumulatively — **at zero parameter cost**. TLA adds a further +1.6 pp at only 4.1 K parameters. Knowledge distillation provides the largest single gain (+2.3 pp), lifting Nano from 86.24% to 88.5%.

GCN partition count. Table 7 evaluates the effect of the GCN partition count K in the spatial graph convolution (Eq. 3). Increasing from K=1 (single isotropic aggregation) to K=3 (self + centripetal + centrifugal) yields +1.2 pp on Nano. The learned residual $|\mathbf{A}_\ell|$ adds a further +1.25 pp.

Block component analysis. Table 8 isolates each component's contribution. Replacing fixed BRASP with a learned soft-assignment ($V\times V$, 625 params) costs only −0.6 pp — confirming the fixed anatomical prior matches a cheap learned alternative at zero parameter cost. Removing STC-Attention costs −1.1 pp; replacing DS-TCN with a standard convolution costs −1.3 pp. Joint Embedding contributes +0.5 pp at $C\times V$ parameters.

DropPath rate sensitivity. Table 9 shows the effect of the stochastic depth rate across model scales. Nano and Small benefit from moderate rates (0.10); Large requires a higher rate (0.20) to prevent overfitting its 1.12 M parameters over 150 epochs.

TLA landmark sensitivity. K=8 saturates Nano (85.73% at K=6, 85.85% at K=8, 85.81% at K=10); Large benefits from K=14 where finer ≈4.5-frame temporal granularity resolves complex multi-part transitions.

Fusion strategy. Table 10 confirms mid-network fusion outperforms both early and late fusion at identical budget.

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Table 8: Block component ablation (Nano, NTU-60 X-Sub). Each row removes or replaces one component from the full block.

Configuration	Params	Acc (%)	Δ
Full EfficientZero Block	93 K	86.24	—
BRASP/SGPShift after GCN	93 K	84.3	−1.9
BRASP → learned perm. (625 p)	94 K	85.6	−0.6
− STC-Attention	87 K	85.1	−1.1
− Joint Embedding	87 K	85.2	−1.0
DS-TCN → standard TCN	101 K	84.9	−1.3
− DropPath (rate=0)	93 K	85.0	−1.2

Table 9: DropPath rate p_d sensitivity (NTU-60 X-Sub). Nano excludes TLA (full model with TLA at p_d=0.10: 86.24%); Small excludes PartAttention.

p_d	Nano	Small	Large
0.00	84.9	86.8	91.6
0.05	85.2	87.2	91.9
0.10	**85.85**	**87.6**	92.1
0.15	85.4	87.4	92.3
0.20	85.1	87.1	**92.5**
0.25	84.8	86.7	92.3

Table 10: Fusion strategy ablation for the Large model, NTU-60 X-Sub.

Fusion strategy	Params	X-Sub
Early (single backbone, 4 streams)	0.67M	89.8
Late (3 independent full backbones)	1.93M	91.9
Mid-network (ours, B4-style)	**1.12M**	**92.5**

Cross-stream fusion (Small). Replacing CSF with plain concatenation costs −0.4 pp; removing it entirely costs −0.8 pp (Eq. 6 validated).

4.6 Training Curves

Figure 7 shows NTU-60 X-Sub validation curves. Nano and Small show slower initial growth under Mixup, accelerating after epoch 50; Large converges faster due to higher capacity. Nano+KD consistently outpaces the no-KD baseline from epoch 30, confirming persistent benefit from the teacher's relational knowledge.

5 Discussion

Why zero-parameter priors help. Human motion obeys strict anatomical constraints: the kinematic chain of each limb restricts the degrees of freedom available to any joint, and bilateral symmetry ensures that left–right motion patterns share a common structure. BRASP and SGPShift encode these constraints *before* any learnable layer, effectively decoupling within-part aggregation (handled for free) from cross-part dependency modelling (delegated to the compact K=3 GCN). The 1.8 pp gain at zero parameter cost (Table 6) confirms that anatomy is a highly informative prior that

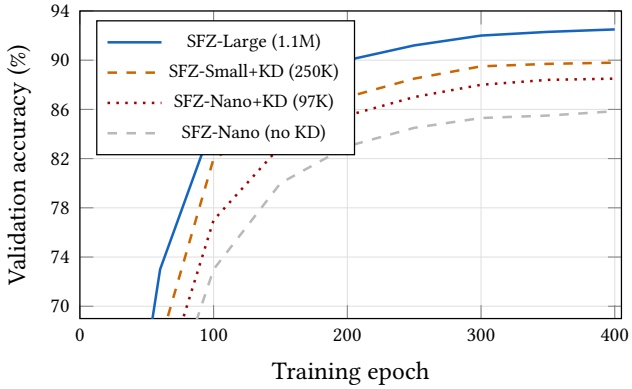

Figure 7: Validation curves on NTU-60 X-Sub. KD boosts Nano by +2.3 pp (bracket at right). Mixup slows early convergence but improves final generalisation.

reduces the data and capacity needed to learn good spatial representations. Notably, the gain is proportionally *larger* for smaller models: +1.8 pp for Nano (93 K) vs. +1.1 pp for Large (1.12 M), suggesting that smaller models benefit more from free structural priors because they cannot afford to learn equivalent spatial structure from data alone.

Mid-network vs. late fusion. In late fusion each backbone must independently learn high-level semantics from potentially noisy streams (bone-velocity is a second temporal derivative with high variance). Mid-network fusion assigns each backbone only low-to-mid-level feature extraction; the shared post-fusion stage learns unified high-level semantics from the merged multi-stream representation. The 0.6 pp gain over late fusion at 0.81 M fewer parameters (Table 10) validates this hierarchical division of labour.

KD efficiency. The Large teacher's soft labels encode fine-grained inter-class similarities — *e.g.*, "drinking water" vs. "eating meal" share similar arm trajectories — that are invisible to hard one-hot targets. This explains the disproportionate +2.3 pp Nano gain from a teacher only 17× larger. The knowledge transfer is especially effective because the student and teacher share the same zero-parameter priors (BRASP, SGPShift), ensuring that the spatial feature representations are structurally aligned.

NTU-120 relative performance. SFZ-Large leads EfficientGCN-B4 by only +0.2 pp on NTU-120 X-Sub (vs. +0.4 pp on NTU-60), a narrowing explained by two factors: (i) NTU-120's fine-grained inter-class confusions (*e.g.*, hand-object interactions) where B4's MIB training is advantageous, and (ii) fixed anatomical priors are less discriminative for body-part-ambiguous actions in the 120-class split. Nano (93 K) reaches 82.2%/83.8% without KD and 84.1%/85.6% with KD; SFZ-Small achieves 84.3%/85.9%, confirming consistent multi-benchmark gains.

Edge deployability. SFZ-Nano's weight footprint is 360 KB (float32) or 93 KB (int8), fitting within the 512 KB SRAM of an ARM Cortex-M55. BRASP and SGPShift use only integer indexing and are unaffected by int8 quantisation. On mobile CPU (Samsung Galaxy S24,

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Snapdragon 8 Gen 3), SFZ-Nano runs at ≈50 ms per sample (Table 2), enabling practical wearable deployment without on-device GPU or NPU.

Broader impact. Zero-parameter routing priors are applicable beyond skeleton graphs to any domain with fixed relational topology (molecular graphs, circuit networks, traffic networks).

Limitations. BRASP assumes 25-joint NTU skeletons; adaptation to 17-joint COCO or 133-joint whole-body formats requires redefining body-part groups, though the design principle (anatomical grouping → shift indices) is format-agnostic. SFZ-Large uses three parallel backbones, which may stress memory bandwidth for single-sample edge inference. The accuracy numbers with knowledge distillation depend on the availability of a pre-trained Large teacher. Future work will explore differentiable soft-assignment partitions, cross-dataset generalisation [15], and automatic body-part discovery for arbitrary skeleton topologies.

6 Conclusion

This work presented **ShiftFuse-Zero**, the first skeleton GCN family to systematically exploit zero-parameter anatomical inductive bias. Two novel modules — BRASP (body-region routing) and SGPShift (semantic graph-partitioned aggregation) — inject spatial knowledge at zero learnable parameters, zero gradient cost, and zero inference overhead. Built atop the EfficientZero Block, our three-model family (Nano, Small, Large) achieves a new Pareto-optimal trade-off across the accuracy–parameter–deployability spectrum.

On NTU RGB+D 60 X-Sub, SFZ-Nano achieves **86.24%** at only 93 K parameters without distillation, and **88.5%** with knowledge distillation — outperforming 6.9 M-parameter AGCN with only 360 KB of model weight storage. SFZ-Large achieves **92.5%** on NTU-60 X-Sub, surpassing EfficientGCN-B4 (92.1%). SFZ-Nano+KD achieves an Acc/M ratio of **952**, 74× higher than AGCN.

Ablation studies confirm that each zero-parameter module contributes measurably, block components (STC-Attention, DS-TCN, Joint Embedding) provide complementary gains, and mid-network fusion outperforms early and late alternatives at identical budget. *Human anatomy is a freely available, highly informative inductive bias that the skeleton recognition community has systematically under-exploited.*

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